UBC ELEC 221: Introduction to Signals and Systems

Continuous Time Signals

Even and Odd Components

$$x_{\text{even}}(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_{\text{odd}}(t) = \frac{1}{2}[x(t) - x(-t)]$$

Periodicity

$$x(t + kT_0) = x(t)\,\forall t \in (-\infty, \infty)$$

Energy

$$E = \int_{-\infty}^{\infty} |x(t)|^2\, dt$$

Power

$$P = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |x(t)|^2\, dt$$

Causality:

- Causal: $x(t) = 0$ for $t < 0$.
- Anti-Causal: $x(t) = 0$ for $t \geq 0$.
- Acausal or Non-Causal: Both of the above.

Continuous Time Systems

Linearity

$$\mathcal{S}[\alpha x(t) + \beta y(t)] = \alpha\mathcal{S}[x(t)] + \beta\mathcal{S}[y(t)]$$

Time Invariance

$$\mathcal{S}[x(t)] = y(t) \Rightarrow \mathcal{S}[x(t \pm \tau)] = y(t \pm \tau)$$

Zero-State Response. Due to the input as the initial conditions are zero.
Zero-Input Response. Due to the initial conditions as the input is zero.
Convolution

$$y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)\, d\tau = \int_{-\infty}^{\infty} x(t-\tau)h(\tau)\, d\tau$$

Causality. A continuous time system $\mathcal{S}$ is causal if whenever $x(t) = 0$ and there are no initial conditions, $y(t) = 0$ and the output $y(t)$ does not depend on future inputs. **BIBO Stability.** If an input bounded then the output of a BIBO stable system is bounded.

$$\Leftrightarrow \int_{-\infty}^{\infty} |h(t)|\, dt < \infty$$

Laplace Transform

Eigenfunction Property

$$\mathcal{S}[e^{s_0 t}] = H(s_0)e^{s_0 t}$$

One Sided Laplace Transform:

$$F(s) = \mathcal{L}[f(t)u(t)] = \int_{0^-}^{\infty} f(t)e^{-st}\, dt, \forall s = \sigma + j\omega$$

Note: $|H(s)|$ is greatest when s is closest to the poles.

Signal Support	**ROC**
Finite support	Entire s-plane.
Causal function	$\sigma > \max(\sigma_i)$, $-\infty < \omega < \infty$
Anti-causal	$\sigma < \min(\sigma_i)$, $-\infty < \omega < \infty$
Non-causal	$\mathcal{R} = \mathcal{R}_{\text{causal}} \cap \mathcal{R}_{\text{anti-causal}}$

Initial Value Theorem

$$f(0^+) = \lim_{s\to\infty} sF(s)$$

Final Value Theorem

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s)$$

BIBO Stability. If the region of convergence contains the $j\omega$-axis then the system is BIBO stable.
2-Sided LT from 1-Sided LT

- For an anti-causal function $C(s) = \mathcal{L}\left[c(-t)u(t)\right]_{(-s)}$
- For a 2-sided function $p(t) = p_{ac}(t) + p_c(t) = p(t)u(-t) + p(t)u(t)$
- $P(t) = \mathcal{L}\left[p_{ac}(-t)u(t)\right]_{(-s)} + \mathcal{L}\left[p_c(t)u(t)\right]$

Fourier Series

Eigenfunction Property

$$\mathcal{S}[e^{j\omega_0 t}] = H(j\omega_0)e^{j\omega_0 t}$$

$$\begin{aligned} x(t) &= \sum_k X_k e^{j\omega_k t} \\ \Rightarrow y(t) &= \sum_k X_k H(j\omega_k)e^{j\omega_k t} \\ &= \sum_k X_k |H(j\omega_k)| e^{j(\omega_k t + \angle H(\omega_k))} \end{aligned}$$

Fourier Series Coefficients (for any t_0)

$$X_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)e^{-jk\omega_0 t}\, dt$$

Parseval's Power Relation (for any t_0):

$$P = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} |x(t)|^2\, dt = \sum_{k=-\infty}^{\infty} |X_k|^2$$

Symmetry of Line Spectra:
$|X_k| = |X_{-k}|$
$\angle X_k = -\angle X_{-k}$
Trigonometric Fourier Series:

$$x(t) = X_0 + 2\sum_{k=1}^{\infty} |X_k| \cos(k\omega t + \Theta_k)$$

$$x(t) = c_0 + 2\sum_{k=1}^{\infty} [c_k \cos(k\omega_0 t) + d_k \sin(k\omega_0 t)]$$

$$c_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)\cos(k\omega_0 t)\, dt \quad k = \mathbf{0}, 1, 2, ..$$

$$d_k = \frac{1}{T_0}\int_{t_0}^{t_0+T_0} x(t)\sin(k\omega_0 t)\, dt \quad k = 1, 2, 3, ...$$

$$\Theta_k = -\arctan(d_k/c_k)$$

Fourier Coefficients from Laplace Transform. If $x_1(t)$ is a single period of $x(t)$, then

$$X_k = \frac{1}{T_0}\mathcal{L}[x_1(t)]\big|_{s=jk\omega_0}$$

Response of LTI Systems to Periodic Signals. If the input to an LTI system has Fourier Series $x(t) = X_0 + 2\sum_{k=1}^{\infty} |X_k| \cos(k\omega t + \angle X_k)$, then the steady state response is

$$y(t) = X_0|H(j0)| + 2\sum_{k=1}^{\infty} |X_k||H(jk\omega_0)| \cos(k\omega_0 t + \angle X_k + \angle H(jk\omega_0))$$

Fourier Transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}\, \mathrm{d}t$$

Inverse Fourier Transform

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)e^{j\omega t}\, \mathrm{d}\omega$$

Fourier Transform from Laplace Transform (if $X(s)$ contains the $j\omega$-axis),

$$\mathcal{F}[x(t)] = \mathcal{L}[x(t)]\big|_{s=j\omega} = X(s)\big|_{s=j\omega}$$

Fourier Transform of Periodic Signals under Fourier Series

$$\sum_k X_k e^{jk\omega_0 t} \Leftrightarrow \sum_k 2\pi X_k \delta(\omega - k\omega_0)$$

Parseval's Energy Relation

$$E = \int_{-\infty}^{\infty} |x(t)|^2\, \mathrm{d}t = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2\, \mathrm{d}\omega$$

Sampling Theory

$$x_s(t) = x(nT_s) = x(t)|_{t=nT_s} = x(t)\sum_n \delta(t - nT_s)$$

$$X_s(\omega) = \frac{1}{T_s}\sum_{k=-\infty}^{\infty} X(\omega - k\omega_s)$$

Nyquist-Shannon Sampling Rate

$$\omega_s = \frac{2\pi}{T_s} \geq 2\omega_{\max}$$

Aliasing occurs if $\omega_s < 2\omega_{\max}$

Reconstruction $X(\omega) = X_s(\omega)H_{\text{lp}}(\omega)$:

$$H_{\text{lp}}(\omega) = \begin{cases} T_s & -\frac{\omega_s}{2} \leq \omega \leq \frac{\omega_s}{2} \\ 0 & \text{otherwise} \end{cases}$$

Inclusive bounds. If represented by a unit step function the bounds are inclusive for filters

Signal Reconstruction from Sinc Interpolation

$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\frac{\sin(\pi(t-nT_s)/T_s)}{\pi(t-nT_s)/T_s}$$

Discrete Time Signals

Definition $x[n] = x(nT_0)$.

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

Periodicity

$$x[n+kN] = x[n] \qquad \forall k \in \mathbb{Z}$$

When sampling an analog sinusoid of fundamental period T_0, we obtain a periodic discrete sinusoid provided that m, N not divisible by each-other,

$$\frac{T_s}{T_0} = \frac{m}{N}$$

Aliasing occurs if

$$T_s > T_0/2$$

Energy

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

Power

$$P_x = \lim_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} |x[n]|^2$$

Convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

Bounded-Input Bounded-Output Stability

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty$$

Z-Transform

$$\mathcal{Z}[x(nT_s)] = \mathcal{L}[x_s(t)]|_{z=e^{sT_s}} = \sum_n x(nT_s)z^{-n}$$

Convergence

$$|X(z)| = \sum_n |x[n]|r^{-n} < \infty$$

Initial Value Theorem

$$x[0] = \lim_{z\to\infty} X(z)$$

Final Value Theorem:

$$\lim_{n\to\infty} x[n] = \lim_{z\to 1}(z-1)X(z)$$

BIBO Stability. If the ROC contains radius $z=1$, then the system is BIBO stable.

Basic Properties of Laplace Transforms

Property	**t-domain**	**s-domain**
Linearity	$\alpha f(t) + \beta g(t)$	$\alpha F(s) + \beta G(s)$
1st Derivative	$f'(t)$	$s\mathrm{F}(s) - f(0^-)$
2nd Derivative	$f''(t)$	$s^2\mathrm{F}(s) - sf(0^-) - f'(0^-)$
Frequency Differentiation	$(-t)^n f(t)$	$\frac{\mathrm{d}^n}{\mathrm{d}s^n}\mathrm{F(s)}$
Integration	$\int_{0^-}^{t} f(\tau)\,\mathrm{d}\tau$	$\frac{\mathrm{F}(s)}{s}$
Frequency Integration	$\frac{f(t)}{t}$	$\int_{-\infty}^{-s} F(-\tau)\mathrm{d}\tau$
Convolution	$\int_0^t f(t-\tau)g(\tau)\,\mathrm{d}\tau$	$\mathrm{G}(s)\mathrm{F}(s)$
Time-shifting	$f(t-\tau)$	$e^{-\tau s}\mathrm{F}(s)$
Frequency-shift	$e^{at}f(t)$	$\mathrm{F}(s-a)$
Time Scaling	$f(at)$	$\frac{1}{a}\mathrm{F}\left(\frac{s}{a}\right)$

One-sided Laplace Transforms

$\mathbf{f(t)}$	$\mathbf{F(s)}$	**Region of Convergence**
$\delta(t)$	e^{-as},	whole s-plane
$u(t)$	$\frac{1}{s}$,	$\mathrm{Re}[s] > 0$
$tu(t)$	$\frac{1}{s^2}$	$\mathrm{Re}[s] > 0$
$t^n u(t)$	$\frac{n!}{s^{n+1}}$	$\mathrm{Re}[s] > 0$
$\sin(\omega t)u(t)$	$\frac{\omega}{s^2+\omega^2}$,	$\mathrm{Re}[s] > 0$
$\cos(\omega t)u(t)$	$\frac{s}{s^2+\omega^2}$,	$\mathrm{Re}[s] > 0$
$t\sin(\omega_0 t)u(t)$	$\frac{2\omega_0 s}{(s^2+\omega_0^2)^2}$,	$\mathrm{Re}[s] > 0$
$t\cos(\omega_0 t)u(t)$	$\frac{s^2-\omega_0^2}{(s^2+\omega_0^2)^2}$,	$\mathrm{Re}[s] > 0$
$e^{-at}u(t)$	$\frac{1}{s+a}$,	$\mathrm{Re}[s] > -a$
te^{-at}	$\frac{1}{(s+a)^2}$,	$\mathrm{Re}[s] > -a$
$t^n e^{-at}u(t)$	$\frac{n!}{(s+a)^{n+1}}$,	$\mathrm{Re}[s] > -a$
$e^{-\alpha t}\sin(\beta t)u(t)$	$\frac{\beta}{(s+\alpha)^2+\beta^2}$,	$\mathrm{Re}[s] > -\alpha$
$e^{-\alpha t}\cos(\beta t)u(t)$	$\frac{s+\alpha}{(s+\alpha)^2+\beta^2}$,	$\mathrm{Re}[s] > -\alpha$

Basic Properties of Fourier Series

Property	**t-domain**	$\boldsymbol{\omega}$**-domain**
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha X_k + \beta Y_k$
Power	$\frac{1}{T}\int_0^T |x(t)|^2\mathrm{d}t$	$\sum_{n=-\infty}^{\infty} |X_k|^2$
Differentiation	$x'(t)$	$jk\omega_0 X_k$
Integration (only if $X_0 = 0$)	$\int_{-\infty}^{t} f(\tau)\,\mathrm{d}\tau$	$\frac{X_k}{jk\omega_0}, k \neq 0$
Time-shifting	$x(t-\alpha)$	$e^{-j\alpha w_0 k}X_k$
Frequency-shift	$e^{j\alpha w_0 t}x(t)$	$X_{k-\alpha}$, $\alpha \in \mathbb{Z}$
Convolution	$z(t) = [x * y](t)$	$X_k Y_k$
Symmetry	x(t) real	$|X_k| = |X_{-k}|$ even $\angle X_k = -\angle X_{-k}$ odd

Basic Properties of CT Fourier Transforms

Property	**t-domain**	$\boldsymbol{\omega}$**-domain**
Linearity	$\alpha x(t) + \beta y(t)$	$\alpha X(\omega) + \beta Y(\omega)$
Time Scaling	$x(\alpha t), \alpha \neq 0$	$\frac{1}{|\alpha|}X\left(\frac{\omega}{\alpha}\right)$
Energy	$\int_{-\infty}^{\infty} |x(t)|^2\mathrm{d}t$	$\frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\omega)|^2\mathrm{d}\omega$
Reflection	$x(-t)$	$X(-\omega)$
Duality	$X(t)$	$2\pi x(-\omega)$
Differentiation	$\frac{\mathrm{d}^n}{\mathrm{d}t^n}x(n)$	$(j\omega)^n X(\omega)$
Frequency Differentiation	$-jtx(t)$	$\frac{\mathrm{d}}{\mathrm{d}\omega}X(\omega)$
Integration	$\int_{-\infty}^{t} x(\tau)\,\mathrm{d}\tau$	$\frac{X(\omega)}{j\omega} + \pi X(0)\delta(\omega)$
Time-shifting	$x(t-\alpha)$	$e^{-j\alpha w}X(\omega)$
Frequency-shift	$e^{jw_0 t}x(t)$	$X(\omega - \omega_0)$
Modulation	$x(t)\cos(\omega_0 t)$	$\frac{1}{2}[X(\omega-\omega_0) + X(\omega+\omega_0)]$
Periodic signals	$\sum X_k e^{jw_0 t}$	$\sum 2\pi X_k \delta(\omega - k\omega_0)$
Symmetry	x(t) real	$|X(\omega)| = |X(-\omega)|$ even $\angle X(\omega) = -\angle X(-\omega)$ odd
Windowing	$x(t)y(t)$	$\frac{1}{2\pi}[X * Y](\omega)$
Cosine Transform	$x(t)$ even	$\int_{-\infty}^{\infty} x(t)\cos(\omega t)\mathrm{d}t$
Sine Transform	$x(t)$ odd	$-j\int_{-\infty}^{\infty} x(t)\sin(\omega t)\mathrm{d}t$

CT Fourier Transform Pair

$\mathbf{x(t)}$	$\mathbf{X}(\omega)$	
1	$2\pi\delta(\omega)$	
$\delta(t)$	$1,$	
$\delta(t-\tau)$	$e^{-j\omega\tau}$	
$u(t)$	$\frac{1}{j\omega}+\pi\delta(\omega)$	
$u(-t)$	$\frac{-1}{j\omega}+\pi\delta(\omega)$	
$\operatorname{sgn}(t)=2[u(t)-0.5]$	$\frac{2}{j\omega}$	
$e^{-at}u(t)$	$\frac{1}{a+j\omega},$	$a>0$
$e^{at}u(-t)$	$\frac{1}{a-j\omega},$	$a>0$
$te^{at}u(t)$	$\frac{1}{(a+j\omega)^2},$	$a>0$
$e^{-a\lvert t\rvert}$	$\frac{2a}{a^2+\omega^2},$	$a>0$
$\cos(\omega_0 t)$	$\pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$	
$\sin(\omega_0 t)$	$-j\pi[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]$	
$u(t+\tau)-u(t-\tau)$	$2\tau\frac{\sin(\omega\tau)}{\omega\tau},$	$\tau>0$
$\frac{\sin(\omega_0 t)}{\pi t}$	$u(\omega+\omega_0)-u(\omega-\omega_0)$	
$\sum_{n=-\infty}^{\infty}\delta(t-nT)$	$\omega_0\sum_{n=-\infty}^{\infty}\delta(\omega-n\omega_0),$	$\omega_0=\frac{2\pi}{T}$

Basic Properties of Z-Transforms

Property	**t-domain**	**z-domain**
Linearity	$\alpha x[n]+\beta y[t]$	$\alpha X(z)+\beta Y(z)$
Convolution	$(x*y)[n]$ $=\sum_k x[k]y[n-k]$	$X(z)Y(z)$
Time-delay (Causal)	$x[n-N]$ $(N\in\mathbb{Z})$	$z^{-N}X(z)$
Time-delay (Causal)	$x[n-N]$ $(N\in\mathbb{Z})$	$z^{-N}X(z)+\sum_{k=1}^{N}x[-k]z^{-N+k}$
Time reversal	$x[-n]$	$X(z^{-1})$
Multiplication by n	$nx[n]$	$-z\frac{\mathrm{d}}{\mathrm{d}z}X(z)$
Multiplication by n^2	$n^2x[n]$	$z^2\frac{\mathrm{d}^2}{\mathrm{d}z^2}X(z)+z\frac{\mathrm{d}}{\mathrm{d}z}X(z)$
Finite Difference	$x[n]-x[n-1]$	$(1-z^{-1})X(z)-x[-1]$
Accumulation	$\sum_{k=0}^{n}x[k]$	$\frac{X(z)}{1-z^{-1}}$

One-sided Z-Transforms

f[n]	**X(z)**	**Region of Convergence**
$\delta[n]$	$1,$	whole z-plane
$u[n]$	$\frac{1}{1-z^{-1}},$	$\lvert z\rvert>1$
$nu[n]$	$\frac{z^{-1}}{(1-z^{-1})^2}$	$\lvert z\rvert>1$
$n^2u[n]$	$\frac{z^{-1}(1+z^{-1})}{(1-z^{-1})^3},$	$\lvert z\rvert>1$
$\alpha^n u[n], \lvert\alpha\rvert<1$	$\frac{1}{1-\alpha z^{-1}},$	$\lvert z\rvert>\lvert\alpha\rvert$
$n\alpha^n u[n], \lvert\alpha\rvert<1$	$\frac{\alpha z^{-1}}{(1-\alpha z^{-1})^2},$	$\lvert z\rvert>\lvert\alpha\rvert$
$\cos(\omega_0 n)u[t]$	$\frac{1-\cos(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}},$	$\lvert z\rvert>1$
$\sin(\omega_0 n)u[t]$	$\frac{\sin(\omega_0)z^{-1}}{1-2\cos(\omega_0)z^{-1}+z^{-2}},$	$\lvert z\rvert>1$
$\alpha^n\cos(\omega_0 n)u[t]$ $\lvert\alpha\rvert<1$	$\frac{1-\alpha\cos(\omega_0)z^{-1}}{1-2\alpha\cos(\omega_0)z^{-1}+\alpha^2z^{-2}},$	$\lvert z\rvert>\lvert\alpha\rvert$
$\alpha^n\sin(\omega_0 n)u[t]$ $\lvert\alpha\rvert<1$	$\frac{\alpha\sin(\omega_0)z^{-1}}{1-2\alpha\cos(\omega_0)z^{-1}+\alpha^2z^{-2}},$	$\lvert z\rvert>\lvert\alpha\rvert$

Two-sided Z Transforms

f[n]	**X(z)**	**Region of Convergence**
$-u[-n-1]$	$\frac{1}{1-z^{-1}},$	$\lvert z\rvert<1$
$-\alpha^n u[-n-1]$	$\frac{1}{1-\alpha z^{-1}},$	$\lvert z\rvert<\lvert\alpha\rvert$
$-n\alpha^n u[-n-1]$	$\frac{1}{1-\alpha z^{-1}}-\frac{1}{1-\alpha^{-1}z^{-1}}$	$\lvert\alpha\rvert<\lvert z\rvert<\frac{1}{\lvert\alpha\rvert}$

Basic Properties of DT Fourier Transforms

Property	**t-domain**	**Ω-domain**
Z-transform	$x[n], X(z),$ $\lvert z\rvert=1\in$ ROC	$X(z=e^{j\Omega})$
Periodicity	$x[n]$	$X(e^{j(\Omega+2\pi k)})$
Time-shifting	$x[n-N]$	$e^{-j\Omega N}X(e^{j\Omega})$
Frequency-shift	$x[n]e^{j\Omega_0 n}$	$X(e^{j(\Omega-\Omega_0)})$
Convolution	$(x*y)[n]$	$X(e^{j\Omega})Y(e^{j\Omega})$
Multiplication	$x[n]y[n]$	$\frac{1}{2\pi}\int_{-\pi}^{\pi}X(e^{j\theta})Y(e^{j(\Omega-\theta)})\mathrm{d}\theta$
Symmetry	$x[n]\in\mathbb{R}$	$\lvert X(e^{j\Omega})\rvert$ even function of Ω $\angle X(e^{j\Omega})$ odd function of Ω
Power	$\sum_{n=-\infty}^{\infty}\lvert x[n]\rvert^2$	$\frac{1}{2\pi}\int_{-\pi}^{\pi}\lvert X(e^{j\Omega})\rvert^2\mathrm{d}\Omega$

Discrete-Time Fourier Transform (DTFT)

$\mathbf{x[n]}$	$\mathbf{X}(e^{j\Omega})$	
$\delta[n]$	1	$\lvert\Omega\rvert<\pi$
$1,$	$2\pi\delta(\Omega)$	$\lvert\Omega\rvert<\pi$
$e^{j\Omega_0 n}$	$2\pi\delta(\Omega-\Omega_0)$	$\lvert\Omega\rvert<\pi$
$\alpha^n u[n], \lvert\alpha\rvert<1$	$\frac{1}{(1-\alpha e^{-j\Omega})^2}$	$\lvert\Omega\rvert<\pi$
$\cos(\omega_0 n)u[n]$	$\pi[\delta(\Omega-\Omega_0)+\delta(\Omega+\Omega_0)]$	$\lvert\Omega\rvert<\pi$
$\sin(\omega_0 n)u[n]$	$\frac{\pi}{j}[\delta(\Omega-\Omega_0)+\delta(\Omega+\Omega_0)]$	$\lvert\Omega\rvert<\pi$
$\alpha^{\lvert n\rvert}, \lvert\alpha\rvert<1$	$\frac{1-\alpha^2}{1-2\cos(\Omega)+\alpha^2}$	$\lvert\Omega\rvert<\pi$
$u[n+\frac{N}{2}]-u[n+\frac{N}{2}]$	$\frac{\sin(\Omega(N+1)/2)}{\sin(\Omega/2)}$	$\lvert\Omega\rvert<\pi$
$\alpha^n\cos(\omega_0 n)u[n]$	$\frac{1-\alpha\cos(\Omega_0)e^{-j\Omega}}{1-2\alpha\cos(\Omega_0)e^{-j\Omega}+\alpha^2e^{-2j\Omega}}$	
$\alpha^n\sin(\omega_0 n)u[n]$	$\frac{\alpha\sin(\Omega_0)e^{-j\Omega}}{1-2\alpha\cos(\Omega_0)e^{-j\Omega}+\alpha^2e^{-2j\Omega}}$	

Linear Algebra

State-space Representation

$$\dot{\mathbf{x}}=A\,\mathbf{x}+B\,\mathbf{u},\mathbf{y}=C\,\mathbf{x}+D\,\mathbf{u}$$

Transfer function matrix

$$\mathbf{H}(s)=(sI-A)^{-1}B+D$$

Diagonalization

$$A=PDP^{-1},\qquad D=\begin{bmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{bmatrix},\qquad P=\begin{bmatrix}\vert & \vert\\ \mathbf{v}_1 & \mathbf{v}_2\\ \vert & \vert\end{bmatrix}$$

Matrix exponential

$$e^{At}=Pe^{Dt}P^{-1},\; e^{Dt}=\begin{bmatrix}e^{\lambda_1 t} & 0\\ 0 & e^{\lambda_2 t}\end{bmatrix}$$

Other Identities

Geometric Series

$$\sum_{k=0}^{n}x^k=\frac{1-x^{n+1}}{1-x},\qquad \sum_{k=0}^{\infty}x^k=\frac{1}{1-x}\qquad(0<x<1)$$

Euler's Formula

$$e^{j\theta}=\cos\theta+j\sin\theta,\quad \cos\theta=\frac{e^{j\theta}+e^{-j\theta}}{2},\quad \sin\theta=\frac{e^{j\theta}-e^{-j\theta}}{2j}$$